

CAP182 SS2

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Part A – Literature Review

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1. First medical publication

Field	Details
Title	Effective hospital readmission prediction models using machine-learned features
Harvard reference	Davis, S., Zhang, J., Lee, I., Rezaei, M., Greiner, R., McAlister, F.A. and Padwal, R. (2022) 'Effective hospital readmission prediction models using machine-learned features', <i>BMC Health Services Research</i> , 22(1), Article 1415. Available at: https://doi.org/10.1186/s12913-022-08748-y
Link	https://link.springer.com/article/10.1186/s12913-022-08748-y
Modelling	The research compared multiple supervised machine learning approaches for predicting hospital readmission including Logistic regression, Random forest, Gradient boosting and other techniques. The researchers focused on checking if automatically learned features could improve prediction accuracy compared to manually engineered features. They used classification metrics like Area Under the Curve (AUC), precision and recall to compare model performance.
Preprocessing techniques	1.Data preprocessing involved cleaning patient records to remove duplicate and inconsistent records 2.Missing values were identified and handled using appropriate imputation techniques. 3.Historical patient admissions, diagnoses and healthcare utilisation records were transformed into predictive features. 4.Feature engineering was performed to create variables describing previous admissions and healthcare interactions. 5.Feature selection techniques were applied to remove irrelevant variables
Datasets used	The authors state that they used linked administrative hospital data covering the period 2011 to 2017 to generate historical features and train the models The study used Electronic Health Record (EHR) datasets containing patient demographic information, admission and discharge records, diagnosis information, treatment histories, and previous readmission outcomes.
Key takeaways	1. Feature engineering improves model performance. 2. Historical healthcare utilisation data is an important predictor of future readmissions. 3. Ensemble models generally outperform individual models when predicting hospital readmissions.

2. Second medical publication

Field	Details
Title	Predictive machine learning model for 30-day hospital readmissions in a tertiary healthcare setting
Harvard Reference	Halac, D., Cocucci, C. and Camerlingo, S. (2025) 'Predictive machine learning model for 30-day hospital readmissions in a tertiary healthcare setting', Bioinformatics Advances, 5(1). Available online https://academic.oup.com/bioinformaticsadvances/article/5/1/vbaf121/8145567
Link	https://academic.oup.com/bioinformaticsadvances/article/5/1/vbaf121/8145567
Modelling	1.The research built three predictive models: Logistic Regression, Random Forest and LightGBM (Light Gradient Boosting Machine). 2.Bayesian optimisation was used to perform hyperparameter tuning and identify the best model configurations. 3.The models were evaluated using discrimination measures such as 4.Area Under the Receiver Operating Characteristic Curve (AUC), calibration analysis and decision curve analysis. Internal validation was performed using 250 bootstrap resamples for reliable performance and to minimise the risk of overfitting.
Preprocessing Techniques	1.Clinical and demographic variables were validated, categorical variables were encoded for machine learning use, and missing values were handled during preprocessing. 2.Feature selection was based on both clinical relevance and statistical significance. 3.Hyperparameter optimisation and calibration analysis formed part of the model development workflow.
Datasets used	Dataset containing 3,388 adult hospital admissions collected from a tertiary healthcare facility. The dataset included demographic variables such as age, clinical indicators such as hypertension, diabetes, heart failure, stroke history, cancer diagnosis, as well as duration variables including length of stay and bedridden status. The target variable was 30-day hospital readmission.
Key takeaways	1. Random Forest performed better than Logistic Regression and LightGBM in predicting readmission risk 2. Chronic conditions and length of stay are strong predictors of readmissions 3. Model validation is important to ensure reliable healthcare predictions

3. Third medical publication

Field	Details
Title	Explainable machine learning for ICU readmission prediction
Harvard reference	de Sá, A.G.C., Gould, D., Fedyukova, A., Nicholas, M., Dockrell, L., Fletcher, C., Pilcher, D., Capurro, D., Ascher, D.B., El-Khawas, K. and Pires, D.E.V. (2023) 'Explainable Machine Learning for ICU Readmission Prediction'. Available on arXiv. [arxiv.org]
Link	https://arxiv.org/html/2309.13781
Modelling	1. The aim was to build machine learning project for Intensive Care Unit (ICU) readmission prediction. 2. A few models were evaluated, and Random Forest achieved the strongest performance with an AUC of approximately 0.70. 3. In addition to predictive modelling, SHAP (Shapley Additive Explanations) was incorporated to provide explainability and interpretability. 4. Likelihood ratio analysis and calibration analysis were also applied to understand the model behaviour and reliability.
Preprocessing techniques	1. The researchers combined data from multiple critical care databases and did preprocessing to ensure consistency. Missing values were identified and treated, clinical variables were standardised, and predictor variables were extracted from structured patient records. 2. Data harmonisation was performed to ensure compatibility between datasets. 3. They also prepared the data for explainability analysis using SHAP values to allow each prediction to be interpreted.
Datasets used	The research used two healthcare datasets: the eICU Collaborative Research Database and the MIMIC-IV Clinical Database. These datasets contain large-scale intensive care unit information including patient demographics, lab test results, admission information, diagnosis data, treatment records and discharge outcomes. The datasets were used to develop and validate ICU readmission prediction models
Key takeaways	1. Random Forest provides a good balance between predictive performance and interpretability 2. Clinical measurements, laboratory results and demographic variables are one of the strongest predictors of patient readmission risk.

4. The public dataset that will be used for Part B

Field	Details
Name	MIMIC-IV Clinical Database
Harvard reference	Johnson, A.E.W., Bulgarelli, L., Pollard, T.J., Horng, S., Celi, L.A. and Mark, R.G. (2024) <i>MIMIC-IV Clinical Database</i> . PhysioNet.
Link	https://physionet.org/content/mimiciv/
Number of instances	The data consists of an estimated 364 000 patients and over 545 000 hospital admissions
Number of feature columns	There are 33 feature columns between the 5 tables I selected for the project. The data consists of a few linked tables. For this project, the tables used are patients, admissions, services, drgcodes, and transfers.
Number of classes	2 Classes: Readmitted within 30 days (Yes) and Not Readmitted within 30 days (No).
Target range	It is not applicable because this is a binary classification problem.
Origin	The dataset originates from Beth Israel Deaconess Medical Center in Boston, Massachusetts, USA. It contains de-identified electronic health records and is distributed through PhysioNet for healthcare research and machine learning applications.